

Packet-to-Cloud Translation Sprint

45-minute individual handwritten activity after networking basics

Name:		Register No.:	
Chosen Individual Title:		Date:	

Activity Mission

You have already studied how a digital message travels through browser, TLS, TCP/IP, NIC, Wi-Fi, routers, internet hops and the destination server. Now convert that networking journey into a cloud architecture for your chosen individual project title. You may use your phone, Google or ChatGPT, but the final answers must be handwritten in your own technical words.

Rules

1	Individual work. Discussing concepts is allowed for 2 minutes only; final answers must be individual.
2	Phone internet, Google and ChatGPT are allowed. Blind copy-paste is not allowed. Write the idea in your own words.
3	Your chosen title must appear in every technical decision: endpoint, service model, cloud component and metric.
4	Use technical terms: DNS, HTTPS/TLS, TCP/IP, routing, API gateway, authentication, database, storage, scaling, monitoring.

45-Minute Flow

Time	Task	Expected Output
0-5 min	Write your project title and identify one main user action.	One clear user action such as login, submit, search, upload or track.
5-15 min	Trace the message journey from phone to cloud.	Network path with protocol/layer terms.
15-30 min	Map the destination into cloud services.	Cloud components, service model and reason.
30-40 min	Design one API and one reliability/security decision.	Endpoint, method, data, authentication, scaling or failure handling.
40-45 min	Self-check and submit.	Readable handwritten sheet with no empty technical boxes.

Part A - Convert Your Title into a Cloud Request

A1. Write your chosen project title. Then write one main user action that must travel from phone/browser to cloud. Example: Submit complaint, upload image, get route, predict disease risk.

A2. Write the exact sentence of the request as if the user tapped a button. Example: 'Send this attendance record to the cloud and show confirmation.'

Part B - Network Journey: Phone to Cloud

Draw and label the path of your chosen request. Use arrows. Minimum labels required: Phone/Browser, DNS, HTTPS/TLS, TCP, IP, Wi-Fi/4G/5G, Router, ISP, Internet, Cloud endpoint, API gateway/server.

B1. Hand-drawn message path with at least 10 correct labels

B2. Fill the technical translation table.

Layer / Component	What it does in your app request	Possible failure	Cloud-side protection
DNS			
HTTPS / TLS			
TCP / IP			
Router / ISP / Internet			
API Gateway / Load Balancer			

Part C - Cloud Landing Zone for Your Title

Cloud Need	Your Component Choice	Service Model	Justification
User identity / login		IaaS / PaaS / SaaS / XaaS	
App backend / business logic		IaaS / PaaS / SaaS / XaaS	
Database / records		IaaS / PaaS / SaaS / XaaS	
File/image storage		IaaS / PaaS / SaaS / XaaS	
Monitoring / analytics		IaaS / PaaS / SaaS / XaaS	

Part D - API Contract and Cloud Behaviour

Design one API endpoint for the user action. It must be cloud-ready, secure and measurable.

Field	Your Handwritten Answer
Endpoint URL path	
HTTP method	GET / POST / PUT / DELETE
Request data	
Response data	
Authentication rule	
Error case	

Part E - Elasticity, Scalability and Measured Service

Mini Numerical Design Problem			
Assume your app receives 7,200 requests in 10 minutes during peak time. One cloud instance can safely handle 180 requests/minute. You must keep 25% spare capacity.	E1. Calculate peak requests/minute.	E2. Calculate required capacity after spare capacity.	E3. Calculate minimum number of instances.

E4. Write one elastic/scalable design decision for your project. Example: auto-scale API servers, cache frequent queries, use managed database read replicas, queue background tasks.

E5. Write two measured-service metrics your cloud dashboard must show. Example: request count, latency, error rate, active users, storage usage, database reads.

Part F - AI/Google Research Evidence

Use your phone to verify one technical choice. Write the source/tool and your own conclusion. Do not copy a paragraph.

Search / Prompt Used	Tool Used	Useful Finding	How I applied it
	Google / ChatGPT / Other		

Evaluation Rubric - 20 Marks

Criteria	Marks	Professor Check
Correct networking path and protocol/layer use	5	
Cloud service model mapping with justification	5	
API contract completeness and security thinking	4	
Elasticity/scalability calculation and design decision	4	
Originality, handwritten clarity and AI/Google synthesis	2	

Submission statement: I used internet/AI only for reference and wrote the final technical answers in my own words.

Student Signature:		Faculty Initial:	
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